

SUZANNA PARKINSON

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EDUCATION

University of Chicago, Chicago, IL

Ph.D., Computational and Applied Mathematics

June 2026

Thesis: “The Role of Depth in Neural Networks: A Representation Cost Perspective”

Advisor: Rebecca Willett, Ph.D.

Brigham Young University, Provo, UT

B.S., Mathematics

August 2020

RESEARCH INTERESTS

Deep learning, neural networks, generalization, optimization, inductive bias, depth separation, single- and multi-index models, bottleneck rank, gradient descent, machine learning, data science.

RESEARCH EXPERIENCE

University of Chicago Physical Sciences Division, Chicago, IL

Graduate Researcher under Professor Rebecca Willett

October 2022 – Present

- Proved that more functions can be learned with polynomial sample complexity using three-layer ReLU networks than with two-layer ReLU networks
- Showed that the size of weights needed to represent functions using neural networks with many linear layers is related to how close the function is to being a multi-index model
- Analyzed how linear neural networks adapt to low-dimensional structure in data when trained via gradient descent with standard hyperparameter settings

Graduate Researcher under Professor Matthew Stephens

June 2021 – December 2022

- Developed algorithms to identify branching structure in data using matrix factorizations

Brigham Young University, Provo, UT

Research Assistant under Professor Tyler Jarvis

March 2018 – August 2020

- Analyzed algorithms for finding zeros of nonlinear systems of equations

Research Assistant under Professor Gus Hart

May 2018 – August 2018

- Derived formulas and wrote Python code for density functional theory calculations

National Security Agency Director’s Summer Program, Fort G. Meade, MD

Intern

May 2019 – August 2019

- Conducted statistical analysis on an implementation of a popular post-quantum cryptographic algorithm
- Submitted detailed findings in an internal refereed technical paper

PUBLICATIONS

In Submission

5. H. Laus, **S. Parkinson**, V. Charisopoulos, F. Krahmer, and R. Willett, “Solving Inverse Problems with Deep Linear Neural Networks: Global Convergence Guarantees for Gradient Descent with Weight Decay,” Dec 2025, *arXiv*: arXiv:2502.15522. doi: [10.48550/arXiv.2502.15522](https://doi.org/10.48550/arXiv.2502.15522).

Peer-Reviewed Publications

4. **S. Parkinson**, G. Ongie, and R. Willett, “ReLU Neural Networks with Linear Layers Are Biased towards Single- and Multi-index Models,” *SIAM Journal on Mathematics of Data Science*, vol. 7, no. 3, pp. 1021–1052, Sep. 2025, doi: [10.1137/24M1672158](https://doi.org/10.1137/24M1672158).
3. J. Blanc, M. Steiner, L. Blake, E. Gibbons, M. Ianni-Ravn, R. Morgan, **S. Parkinson**, C. Porras, and E. Zhong, “Ten simple rules for success as a trainee-led outreach organization in computational biology education,” *PLOS Computational Biology*, vol. 21, no. 7, pp. 1–12, Jul. 2025, doi: [10.1371/journal.pcbi.1013281](https://doi.org/10.1371/journal.pcbi.1013281).

2. **S. Parkinson**, G. Ongie, R. Willett, O. Shamir, and N. Srebro, “Depth Separation in Norm-Bounded Infinite-Width Neural Networks,” in *Proceedings of Thirty Seventh Conference on Learning Theory*, S. Agrawal and A. Roth, Eds., in Proceedings of Machine Learning Research, vol. 247. PMLR, Jul. 2024, pp. 4082–4114. [Online]. Available: <https://proceedings.mlr.press/v247/parkinson24a.html>
1. **S. Parkinson**, H. Ringer, K. Wall, E. Parkinson, L. Erekson, D. Christensen, T. Jarvis, “Analysis of normal-form algorithms for solving systems of polynomial equations,” *Journal of Computational and Applied Mathematics*, vol. 411, p. 114235, Sep. 2022, doi: [10.1016/j.cam.2022.114235](https://doi.org/10.1016/j.cam.2022.114235).

HONORS & AWARDS

<i>American Dissertation Fellowship</i> , American Association of University Women	2025-2026
<i>Graduate Research Fellowship</i> , National Science Foundation	2020-2025
<i>Student Travel Award: MDS2024</i> , Society for Industrial and Applied Mathematics	2024
<i>Outstanding Poster Award</i> , University of Chicago Statistics Department Student Poster Day	2024
<i>Outstanding Teaching Assistant Award</i> , University of Chicago Computational and Applied Mathematics	2022
<i>Elaine K. Bernstein Fellowship for Women in Science</i> , University of Chicago	2020
<i>Junior of the Year</i> , Brigham Young University Mathematics Department	2019
<i>Outstanding Poster Award</i> , Joint Mathematics Meetings	2019
<i>Presidential Scholarship</i> (1.5x Tuition), Brigham Young University	2015-2020

PRESENTATIONS

Seminars & Invited Talks

16. **S. Parkinson** (2025). *On The Role of Depth in Deep Learning*, Brigham Young University Mathematics Colloquium, Provo, UT
15. **S. Parkinson** (2025). *On The Role of Depth in Deep Learning*, John Reinitz Memorial Lecture, University of Chicago Computational & Applied Mathematics, Chicago, IL
14. **S. Parkinson** (2025). *Neural Networks Can Automatically Adapt to Low-Dimensional Structure in Inverse Problems*, Brigham Young University Applied Analysis Seminar, Provo, UT
13. **S. Parkinson** (2024). *Depth Separation in Learning via Representation Costs*, Brigham Young University Applied Math Seminar, Provo, UT

Conferences & Workshops

12. **S. Parkinson** (2025). *Depth Separation in Learning via Representation Costs*, IFDS Workshop on Theoretical Foundations of Applied AI, Seattle, Washington
11. **S. Parkinson** (2024). *Linear Layers in ReLU Networks Promote Learning Single-/Multiple-Index Models*, SIAM Conference on Mathematics of Data Science, Atlanta, GA
10. **S. Parkinson** (2024). *Depth Separation in Learning via Representation Costs*, Computational Harmonic Analysis in Data Science and Machine Learning, Oaxaca, Mexico
9. **S. Parkinson**, S. (2023). *Finding Low-Rank Functions Using Linear Layers in Neural Networks*, University of Chicago Computational and Applied Mathematics Student Seminar, Chicago, IL
8. L. Erekson, **S. Parkinson**, D. Christensen, N. Larsen, T. Jarvis (2020). *A Hybrid Multivariate Root-finding Method For Smooth Functions*, Joint Mathematics Meetings, Denver, CO
7. **S. Parkinson**, N. Larsen, E. Parkinson, H. Ringer, T. Moncur, T. Jarvis (2019). *Fast, stable multivariate numerical rootfinding in a compact region.*, Joint Mathematics Meetings, Baltimore, MD

Posters

6. **S. Parkinson** (2025). *Depth Separation in Learning via Representation Costs*, IFDS Workshop on Theoretical Foundations of Applied AI, Seattle, Washington
5. **S. Parkinson** (2025). *Depth Separation in Learning via Representation Costs*, Midwest Machine Learning Symposium, Chicago, IL
4. **S. Parkinson** (2024). *Depth Separation in Learning via Representation Costs*, Conference on Learning Theory, Edmonton, Canada
3. **S. Parkinson** (2024). *Depth Separation in Learning via Representation Costs*, University of Chicago Statistics Department Student Poster Day, Chicago, IL
2. **S. Parkinson**, S. (2023). *Linear Layers Promote Learning Single-/Multiple-Index Models*, Midwest Machine Learning Symposium, Chicago, IL
1. **S. Parkinson**, N. Larsen, E. Parkinson, H. Ringer, T. Moncur, T. Jarvis (2019). *Numerical rootfinding on a compact region.*, Joint Mathematics Meetings, Baltimore, MD

TEACHING EXPERIENCE

University of Chicago, Chicago, IL

Teaching Assistant, Data Science Institute

September 2025

- Developed and presented hands-on coding notebooks to illustrate concepts and introduce technical tools
- Schmidt AI in Science Deep Dive for Postdoctoral Researchers (4-day intensive)

Scientific Content Committee Chair, Computational Biology Outreach Group

October 2022 – January 2024

- Led a committee of 2-6 trainees to develop materials for teaching computational biology to K-12 students
- Prepared teaching materials for 7 events (workshops/science fair booths); presented at 6

Teaching Assistant, Committee on Computational and Applied Mathematics

September 2021 – May 2022

- Graded assignments; answered student questions during office hours and in online discussion groups
- Scientific Computing with Python (2 quarters) & Machine Learning (1 quarter)

University of Washington, Seattle, WA

Teaching Assistant, Department of Biostatistics

July 2022

- Helped students implement computational solutions in Zoom breakout rooms during virtual course
- Monte Carlo Markov Chains for Genetics (1-week bootcamp)

PROFESSIONAL ACTIVITIES

Technical Paper Reviewer

Neural Information Processing Systems (NeurIPS)

2025

SIAM Journal on Mathematics of Data Science (SIMODS)

2025

Conference Special Session Organizer

“Learning Functions with Low-Dimensional Structure using Neural Networks.” SIAM Conference on Mathematics of Data Science, 2024. Co-organized with R. Willett and G. Ongie.